

Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

You have the entire period.

Signature

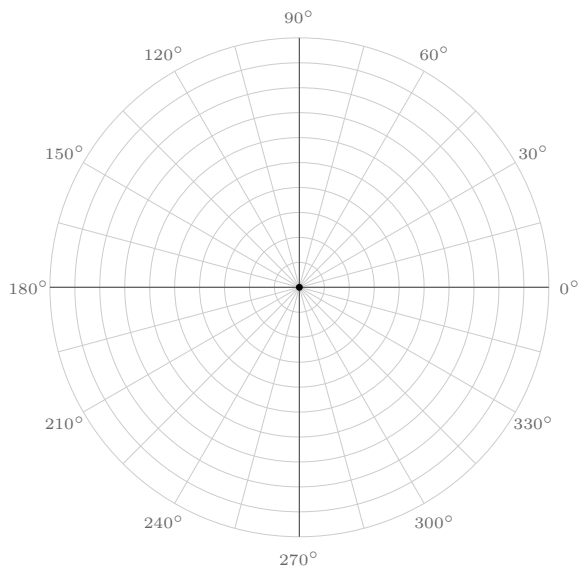
SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator).

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. (3 points each)

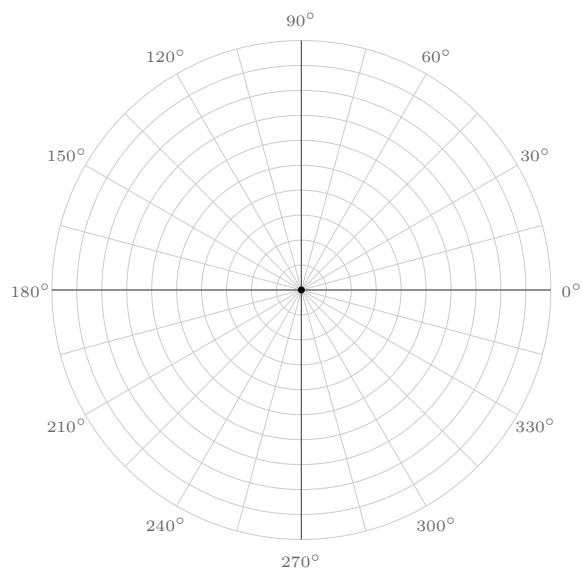
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 3 + 6 \cos \theta$



Type: _____

2. $r^2 = 25 \sin 2\theta$



Type: _____

*Non-Calculator***SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

3.

a) $\sqrt{-36}$

b) 7

Find the product. Write your answer in standard form. (2 points)

4. $\sqrt{-12} \cdot \sqrt{-27}$

Simplify each power of i . (1 point each)

5.

a) i^{38}

b) $\frac{1}{i^{23}}$

Find the product. Write your answer in standard form. (2 points)

6. $(4 - 3i)(-1 + 5i)$

Find the quotient. Write your answer in standard form. (3 points)

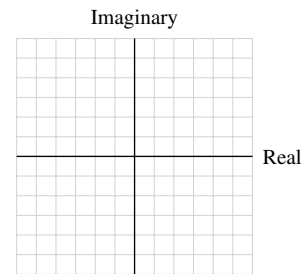
7. $\frac{5 + i}{3 - 2i}$

Solve the equation over the set of complex numbers. (3 points)

8. $2x^2 + 5 = -2x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (2 points)

9. $4 - 3i$, $-1 + 5i$



Write the complex number in rectangular form. (2 points)

10. $6(\cos 150^\circ + i \sin 150^\circ)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

11.

a) $2 - 2\sqrt{3}i$

b) $-7i$

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. Determine whether each number belongs to the Julia set. (1 point each)

12.

a) $z = 0$

YES / NO

b) $z = 1 + i$

YES / NO

Find the product. Write the product in rectangular form, using exact values. (3 points)

13. $[\sqrt{3}(\cos 145^\circ + i \sin 145^\circ)] [\sqrt{3}(\cos 185^\circ + i \sin 185^\circ)]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

14. $\frac{-4\sqrt{3} - 4i}{1 + i\sqrt{3}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(1 + i\sqrt{3})^5$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

16. $x^3 + 27 = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

a) $z = -1$

YES / NO

b) $z = 1$

YES / NO

18. For the point $(-6, 300^\circ)$ in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is **not** an equivalent point, and (c) find the rectangular coordinates of the point in exact form. 3 pts

a) $(6, 120^\circ)$

b) $(-6, -60^\circ)$

c) $(6, -240^\circ)$

d) $(6, 300^\circ)$

19. For the rectangular coordinates $(-2\sqrt{3}, -2)$, give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$. 3 pts

Convert each rectangular equation to a polar equation. (2 points each)

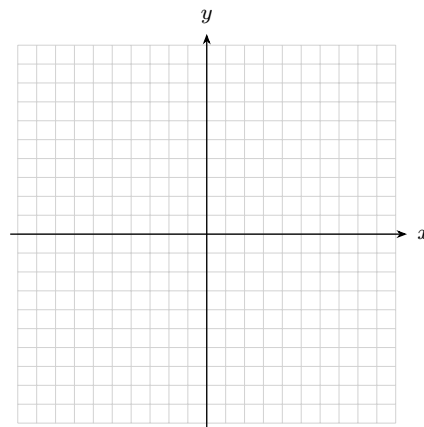
20.

a) $3x - 2y = 6$

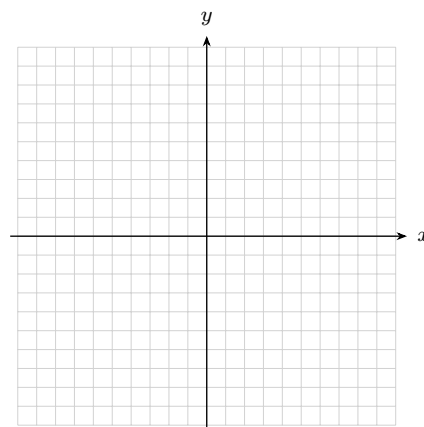
b) $x^2 + y^2 = 49$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

21. $r = 6 \cos \theta$



22. $r = \frac{4}{1 + \sin \theta}$



23. Find the polar equation of the line that passes through $(6, 0^\circ)$ and $(4, 90^\circ)$. Simplify your answer.

3 pts

Find the polar coordinates of the points of intersection of the given curves for the specified interval of θ . (4 points)

24. $r = 2 + 2 \cos \theta$, $r = 3$, $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

a) $r = 5 \sin 3\theta$ _____

b) $r = 4 + 4 \cos \theta$ _____

c) $r^2 = 36 \cos 2\theta$ _____

d) $r = 2 + 5 \sin \theta$ _____

e) $r = 7$ _____

ANSWER KEY – Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

1. $r = 3 + 6 \cos \theta$

WORK

Polar mode: $r1(\theta) = 3 + 6 \cos \theta$

$\theta = 0^\circ: r = 3 + 6(1) = 9$

$\theta = 90^\circ, 270^\circ: r = 3$

$\theta = 180^\circ: r = 3 - 6 = -3$

$\frac{a}{b} = \frac{3}{6} = \frac{1}{2} < 1$

limaçon (inner loop)

WHY

Degree or radian mode both work; a blank screen means the mode or window is off.

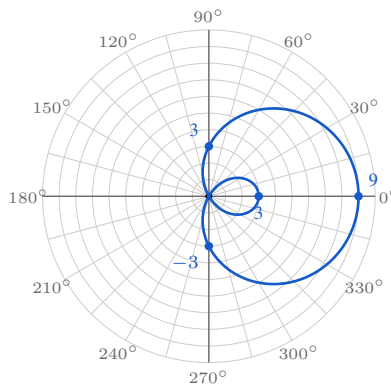
Graph Trace reads r . Rightmost point $(9, 0)$.

Crosses the y -axis at 3 and -3 .

Negative r walks backward: inner loop ends at $(3, 0)$.

Ratio under 1: loop inside.

Form $a + b \cos \theta$ with $a \neq b$.



2. $r^2 = 25 \sin 2\theta$

WORK

$r = \pm \sqrt{25 \sin 2\theta}$

$r1(\theta) = \sqrt{25 \sin(2\theta)}$

$\theta = 45^\circ: r^2 = 25(1), r = 5$

$\sin 2\theta < 0$ in QII and QIV

lemniscate

WHY

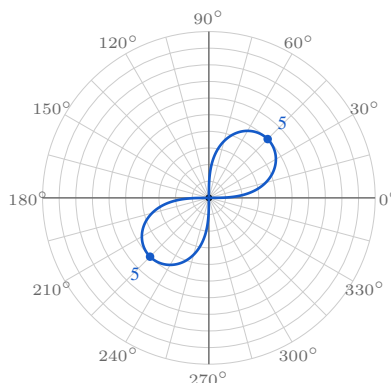
The calculator needs $r =$; either root gives the same graph.

Enter this one; the $-$ root retraces it.

Tip of the loop in QI; 225° gives the QIII tip.

No graph there: loops only in QI and QIII.

r^2 equation, infinity shape.



ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

3.

WORK

$$a) \sqrt{-36} = 6i = 0 + 6i$$

complex, pure imaginary, nonreal complex

$$b) 7 = 7 + 0i$$

real, complex

WHY

$$a = 0, b \neq 0.$$

Every number is complex; $b \neq 0$ makes it nonreal.

$$b = 0.$$

Real numbers are complex too.

Find the product. Write your answer in standard form. (2 points)

$$4. \sqrt{-12} \cdot \sqrt{-27}$$

WORK

$$\sqrt{-12} = 2i\sqrt{3}, \sqrt{-27} = 3i\sqrt{3}$$

$$(2i\sqrt{3})(3i\sqrt{3}) = 6i^2(3)$$

$$= 18i^2 = \boxed{-18}$$

Trap: $\sqrt{(-12)(-27)} = \sqrt{324} = 18$ has the wrong sign.

WHY

Pull out i first.

Multiply the pieces.

$$i^2 = -1.$$

Simplify each power of i . (1 point each)

5.

WORK

$$a) 38 \div 4 = 9 \text{ R } 2, i^{38} = i^2 = \boxed{-1}$$

$$b) 23 \div 4 = 5 \text{ R } 3, i^{23} = i^3 = -i$$

$$\frac{1}{-i} \cdot \frac{i}{i} = \frac{i}{-i^2} = \frac{i}{1} = \boxed{i}$$

WHY

Only the remainder matters.

Remainder 3 means $-i$.

Clear i from the bottom.

Find the product. Write your answer in standard form. (2 points)

$$6. (4 - 3i)(-1 + 5i)$$

WORK

$$-4 + 20i + 3i - 15i^2$$

$$= -4 + 23i + 15$$

$$= \boxed{11 + 23i}$$

WHY

FOIL.

$$-15i^2 = +15.$$

Combine the reals.

Find the quotient. Write your answer in standard form. (3 points)

7. $\frac{5+i}{3-2i}$

WORK

$$\frac{5+i}{3-2i} \cdot \frac{3+2i}{3+2i}$$

top: $15 + 10i + 3i + 2i^2 = 13 + 13i$

bottom: $9 + 4 = 13$

$$\frac{13 + 13i}{13} = \boxed{1 + i}$$

WHY

Multiply by the conjugate of the bottom.

$$2i^2 = -2.$$

$$(a - bi)(a + bi) = a^2 + b^2.$$

Split and reduce.

Solve the equation over the set of complex numbers. (3 points)

8. $2x^2 + 5 = -2x$

WORK

$$2x^2 + 2x + 5 = 0$$

$$x = \frac{-2 \pm \sqrt{4 - 4(2)(5)}}{2(2)} = \frac{-2 \pm \sqrt{-36}}{4}$$

$$= \frac{-2 \pm 6i}{4}$$

$$\boxed{x = -\frac{1}{2} \pm \frac{3}{2}i}$$

WHY

Zero on one side first.

Quadratic formula, $a = 2$, $b = 2$, $c = 5$.

$$\sqrt{-36} = 6i.$$

Split into standard form.

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (2 points)

9. $4 - 3i$, $-1 + 5i$

WORK

$$(4 - 1) + (-3 + 5)i = \boxed{3 + 2i}$$

$$|3 + 2i| = \sqrt{3^2 + 2^2} = \boxed{\sqrt{13}}$$

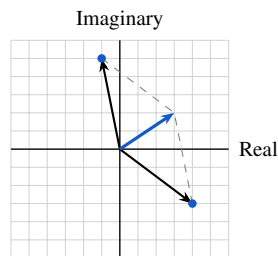
Graph: $(4, -3)$, $(-1, 5)$, resultant $(3, 2)$

WHY

Add reals, add imaginaries.

Absolute value is the distance from 0.

Resultant is the diagonal of the parallelogram.



Write the complex number in rectangular form. (2 points)

10. $6(\cos 150^\circ + i \sin 150^\circ)$

WORK

$$6\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$$

$$= \boxed{-3\sqrt{3} + 3i}$$

WHY

150° is QII: cos negative, sin positive.

Distribute the 6.

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

11.

WORK

$$\text{a) } r = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\theta = \tan^{-1}\left(\frac{-2\sqrt{3}}{2}\right) = \tan^{-1}(-\sqrt{3}) = -60^\circ$$

$$\theta = -60^\circ + 360^\circ = 300^\circ$$

$$\boxed{4 \operatorname{cis} 300^\circ}$$

$$\text{b) } -7i = 0 - 7i: r = 7, \theta = 270^\circ$$

$$\boxed{7 \operatorname{cis} 270^\circ}$$

WHY

$$r = \sqrt{x^2 + y^2}.$$

$\tan^{-1}$ only gives QI or QIV.

$x > 0, y < 0$ is QIV: add 360° .

$r \operatorname{cis} \theta$.

On the negative imaginary axis.

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. Determine whether each number belongs to the Julia set. (1 point each)

12.

WORK

$$\text{a) } 0^2 - 1 = -1$$

$$(-1)^2 - 1 = 0, \text{ then } -1, 0, -1, \dots$$

YES

$$\text{b) } (1 + i)^2 - 1 = 2i - 1 = -1 + 2i$$

$$|-1 + 2i| = \sqrt{1 + 4} = \sqrt{5} \approx 2.24 > 2$$

NO

WHY

$$|-1| = 1.$$

Repeats; never over 2.

$$(1 + i)^2 = 1 + 2i + i^2 = 2i.$$

Exceeds 2 on the first step.

Find the product. Write the product in rectangular form, using exact values. (3 points)

$$\text{13. } [\sqrt{3}(\cos 145^\circ + i \sin 145^\circ)] [\sqrt{3}(\cos 185^\circ + i \sin 185^\circ)]$$

WORK

$$\sqrt{3} \cdot \sqrt{3} \operatorname{cis}(145^\circ + 185^\circ)$$

$$= 3 \operatorname{cis} 330^\circ$$

$$= 3 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= \boxed{\frac{3\sqrt{3}}{2} - \frac{3}{2}i}$$

WHY

Product theorem: multiply the r 's, add the θ 's.

Only the sum has to be a unit-circle angle.

330° is QIV: sin negative.

Distribute.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

14. $\frac{-4\sqrt{3} - 4i}{1 + i\sqrt{3}}$

WORK

$$r_1 = \sqrt{48 + 16} = 8$$

$$\theta_1 = \tan^{-1}\left(\frac{-4}{-4\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$$

$$\theta_1 = 30^\circ + 180^\circ = 210^\circ$$

$$r_2 = \sqrt{1 + 3} = 2, \theta_2 = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$\frac{8 \operatorname{cis} 210^\circ}{2 \operatorname{cis} 60^\circ} = 4 \operatorname{cis} 150^\circ$$

$$= 4 \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \boxed{-2\sqrt{3} + 2i}$$

WHY

Top first.

Calculator-style answer is QI.

$x < 0, y < 0$ is QIII: add 180° .

Bottom is QI, no fix.

Quotient theorem: divide the r 's, subtract the θ 's.

150° is QII.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(1 + i\sqrt{3})^5$

WORK

$$r = \sqrt{1 + 3} = 2, \theta = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$(2 \operatorname{cis} 60^\circ)^5 = 2^5 \operatorname{cis}(5 \cdot 60^\circ)$$

$$= 32 \operatorname{cis} 300^\circ$$

$$= 32 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \boxed{16 - 16\sqrt{3}i}$$

WHY

QI, no fix.

De Moivre: power on r , multiply θ .

300° is QIV.

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

16. $x^3 + 27 = 0$

WORK

$$x^3 = -27 = 27 \operatorname{cis} 180^\circ$$

$$\sqrt[3]{27} = 3$$

$$\alpha = \frac{180^\circ + 360^\circ k}{3} = 60^\circ + 120^\circ k$$

$$3 \operatorname{cis} 60^\circ, 3 \operatorname{cis} 180^\circ, 3 \operatorname{cis} 300^\circ$$

$$\boxed{\frac{3}{2} + \frac{3\sqrt{3}}{2}i, -3, \frac{3}{2} - \frac{3\sqrt{3}}{2}i}$$

WHY

-27 sits on the negative real axis.

n th root theorem: root of r .

$k = 0, 1, 2$: three roots, 120° apart.

Unit circle values times 3.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

WORK

a) $-1, (-1)^2 + (-1) = 0, 0^2 + (-1) = -1$

YES

b) $1, 1^2 + 1 = 2, 2^2 + 1 = 5$

NO

Trap: 2 does not exceed 2. Keep going one more step.

WHY

Back to -1 : it cycles $-1, 0, -1, \dots$

No absolute value ever exceeds 2.

$|5| = 5 > 2$.

18. For the point $(-6, 300^\circ)$ in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is **not** an equivalent point, and (c) find the rectangular coordinates of the point in exact form.

WORK

(a) face 300° (QIV), walk backward: QII

(b) $(-6, 300^\circ) = (6, 300^\circ - 180^\circ) = (6, 120^\circ)$

$(6, -240^\circ)$: same as $(6, 120^\circ)$; $(-6, -60^\circ)$: same as $(-6, 300^\circ)$

$(6, 300^\circ)$ is in QIV: d

(c) $x = -6 \cos 300^\circ = -6 \left(\frac{1}{2}\right) = -3$

$y = -6 \sin 300^\circ = -6 \left(-\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$

$(-3, 3\sqrt{3})$

WHY

Negative r lands in the opposite quadrant.

$(-r, \theta) = (r, \theta \pm 180^\circ)$.

$\pm 360^\circ$ keeps the point.

$x = r \cos \theta$; keep r 's sign.

$y = r \sin \theta$.

QII, matches (a).

19. For the rectangular coordinates $(-2\sqrt{3}, -2)$, give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$.

WORK

$r = \sqrt{(-2\sqrt{3})^2 + (-2)^2} = \sqrt{12 + 4} = 4$

$\tan \theta = \frac{-2}{-2\sqrt{3}} = \frac{1}{\sqrt{3}}, \theta = 30^\circ$

$\theta = 30^\circ + 180^\circ = 210^\circ$

$(4, 210^\circ)$, $(-4, 30^\circ)$

WHY

$r^2 = x^2 + y^2$.

Reference angle.

$x < 0, y < 0$: QIII.

Second pair: $(-r, \theta - 180^\circ)$.

Convert each rectangular equation to a polar equation. (2 points each)

20.

WORK

a) $3r \cos \theta - 2r \sin \theta = 6$

$r(3 \cos \theta - 2 \sin \theta) = 6$

$r = \frac{6}{3 \cos \theta - 2 \sin \theta}$

b) $r^2 \cos^2 \theta + r^2 \sin^2 \theta = 49$

$r^2(1) = 49, r = \pm\sqrt{49}$

$r = 7$ or $r = -7$

WHY

$x = r \cos \theta, y = r \sin \theta$.

Factor out r .

Solve for r .

Substitute.

$\cos^2 \theta + \sin^2 \theta = 1$; $\pm$ on the square root.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

21. $r = 6 \cos \theta$

WORK

$$r^2 = 6r \cos \theta$$

$$x^2 + y^2 = 6x$$

$$x^2 - 6x + 9 + y^2 = 9$$

$$(x - 3)^2 + y^2 = 9$$

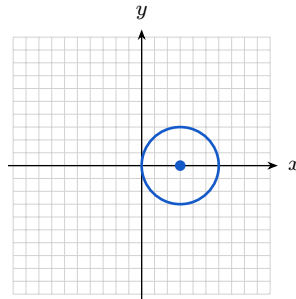
WHY

Multiply both sides by r to make r^2 and $r \cos \theta$.

$$r^2 = x^2 + y^2, r \cos \theta = x.$$

Complete the square: add $\left(\frac{6}{2}\right)^2$ both sides.

Circle, center $(3, 0)$, radius 3.



22. $r = \frac{4}{1 + \sin \theta}$

WORK

$$r(1 + \sin \theta) = 4$$

$$r + r \sin \theta = 4, r + y = 4$$

$$r = 4 - y, \sqrt{x^2 + y^2} = 4 - y$$

$$x^2 + y^2 = 16 - 8y + y^2$$

$$x^2 = -8(y - 2)$$

WHY

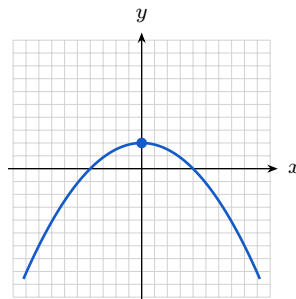
Clear the fraction.

$$r \sin \theta = y.$$

Isolate r , then $r = \sqrt{x^2 + y^2}$.

Square both sides.

Parabola, vertex $(0, 2)$, opens down.



23. Find the polar equation of the line that passes through $(6, 0^\circ)$ and $(4, 90^\circ)$. Simplify your answer.

WORK

$$(6, 0^\circ): x = 6 \cos 0^\circ = 6, y = 6 \sin 0^\circ = 0, (6, 0)$$

$$(4, 90^\circ): x = 0, y = 4, (0, 4)$$

$$m = \frac{4 - 0}{0 - 6} = -\frac{2}{3}, y = -\frac{2}{3}x + 4$$

$$3y = -2x + 12, 2x + 3y = 12$$

$$2r \cos \theta + 3r \sin \theta = 12$$

$$r = \frac{12}{2 \cos \theta + 3 \sin \theta}$$

WHY

Convert both points first.

Slope, then y -intercept 4.

Clear the fraction; standard form.

Substitute.

Factor r , divide.

Find the polar coordinates of the points of intersection of the given curves for the specified interval of θ . (4 points)

24. $r = 2 + 2 \cos \theta$, $r = 3$, $0 \leq \theta < 2\pi$

WORK

$$2 + 2 \cos \theta = 3$$

$$2 \cos \theta = 1, \cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$r = 3$ at both

$$\left(3, \frac{\pi}{3}\right), \left(3, \frac{5\pi}{3}\right)$$

WHY

Set $r = r$.

Isolate the trig function.

Cosine positive: QI and QIV.

Plug back in.

$r = 3$ never reaches the pole, so no pole point.

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

WORK

a)

b)

c)

d)

e)

WHY

$n = 3$ odd: 3 petals, each 5 long.

$a + b \cos \theta$ with $\frac{a}{b} = \frac{4}{4} = 1$.

r^2 equation.

$\frac{a}{b} = \frac{2}{5} \neq 1$ (inner loop).

No θ : every point 7 from the pole.